MHC in DC Independent Study Guide

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Contents

Welcome!

This is an online guide for Mount Holyoke students undertaking an independent study (POLIT 395) while spending the fall semester living, learning, and interning in Washington, DC.

As an integral part of the DC program, the independent study offers you the opportunity to take a deep dive into a political science-related topic of your choice. This guide offers step-by-step advice to assist you as you develop your project from an initial idea to a polished research paper.

What's in this guide?

This guide begins by providing you with the **big picture** of the research process. Like researchers in other social science disciplines, such as economics or sociology, political scientists ultimately want to **explain** aspects of the social and political world. What sets political scientists apart from journalists, pundits, or other political commentators is our commitment to transparency. When it comes to social science research, it's not just *what* you found that's important but also *how* you found it.

Additionally, while understanding of the political world is valuable for its own sake, an underlying motivation for most political scientists is to understand the world *in order to make it better*. Thus, a common focus of most social science is **problem-oriented research**: our work begins with a problem or puzzle of general concern which requires better understanding if society is to respond effectively.

Fortunately, we don't need to tackle these big problems on our own. Social scientists undertake research within communities of likeminded scholars who research similar or sometimes the same problem or puzzle as us. By **joining these ongoing conversations**, we benefit from generations of past and contemporary scholars, inheriting the accumulated knowledge of the discipline, and drawing inspiration and ideas from it as we make our own modest contribution.

Specifically, despite the apparent novelties of things like Artificial Intelligence (AI) or the presidency of Donald Trump, very few things in our world are actually brand new. Though it may be true that AI is a very distinctive kind of technology, and that there's never been an American president quite like Donald Trump, they are both instances of broader phenomena: disruptive innovations on the one hand; right-wing populists on the other. What's useful about looking at new developments in this way is that there is usually some existing **theory** to draw on when trying to explain them. By turning to the existing literature, we can learn how to think abstractly about, say, how disruptive innovations are likely to affect wealth disparities, or about how populist presidents are likely to challenge institutional checks and balances. Whether we are applying existing theory to new problems or creating a new theory from scratch, thinking abstractly about concepts and their relationships provides us with a preliminary understanding of novel people, processes, and events.

To see if a theory actually helps us explain the political problems that concern us, we need to draw out some empirical implications, or **hypotheses**, to test. To do so, we simply ask ourselves:

> *if this theory is true, what would I expect to see in the world?*

Our strategy for **collecting the evidence or data** we need to test our hypotheses is called our **research design**. Specifying precisely how we collected our data and tested our hypotheses is a critical part of research because it is a process containing myriad decisions on the part of the researcher—some small, others large—none of answers for which are obvious and all of which a critical reader may scrutinize. If, say, you're interested in explaining the affect of AI on inequality, how will you define and measure those two concepts? There are multiple legitimate ways to do so, so why did you do it in *that* particular way? Again, it's not just what you found that matters in social science, it's *how* you found it.

As this brief summary implies, social science research takes a significant amount of effort and hard work. But the rewards can be personally edifying and professionally rewarding. The **findings and implications** of our research can change the way we, and maybe even others, understand and intervene in the world. We may even be able to make it better.

How to use this guide

This guide is meant to serve as a learning resource and reference material for you as you chart the path of your independent study from start to finish. In it you'll find chapters that progressively walk you through the research workflow from begging to end, ranging from ideas about how to get started to detailed instructions about how to format your final paper.

The chapters are accompanied by short asynchronous lectures posted to Canvas. These are meant to clarify and amplify specific parts of the readings, not replace them, so be sure to consult both resources throughout the semester.

On Canvas you'll also find links to upload your work. Details regarding due dates can be found in the course syllabus.

1. Introduction: The Big Picture

i Chapter learning goals

- Consider the logic of social science as inference to the best explanation
- Appreciate the strengths and limitations of social science research
- Develop an overview of the research project workflow

This chapter provides an overview of the big picture: that is, it seeks to provide you with a sense of what social science is, how to go about it, and what we can reasonably expect to learn from it. The chapters that follow address specific steps, stages, and strategies of the research process. But in order to see how they all fit together, we begin with the underlying logic of inquiry that animates them.

1.1 The logic of social science: inference to the best explanation

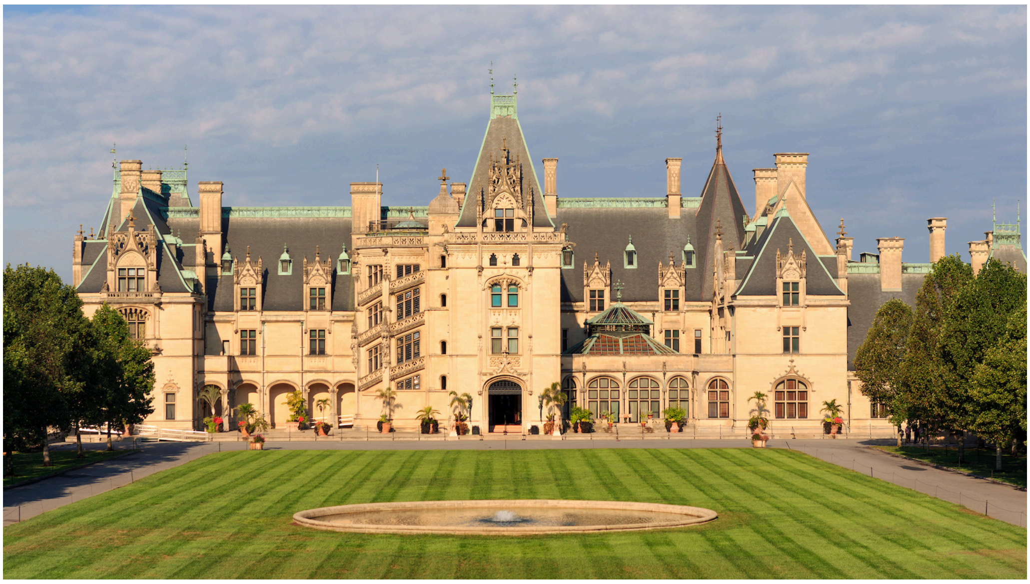

Figure 1.1: Social science, a mansion with many rooms

The broad field of knowledge known as social science is like a mansion with many rooms. Inside each rooms are groups of researchers concerned with a particular type of problem or puzzle. *Why are some nations rich and others poor? Why do some economies industrialize while others stagnate?* Neighboring rooms contain specialists concerned with similar problems. *Why are most stable democracies wealthy while poor democracies often succumb to autocrats?* Rooms down the hall or in another wing of the building are consumed with questions that may seem largely unrelated. *If the US raised the federal minimum wage to $20 per hour, what would happen to the unemployment rate? How has the declining child vaccination rate impacted US public health?*

Despite the rich complexity and varied differences that constitute the field, most social scientists agree on one thing: that the goal of social science is to **explain the social and political world**. It's that simple. The world presents us with problems or puzzles that we can't but would like to explain, so we go about trying to explain them.

1.1.1 Social scientific reasoning

Social scientists have a distinctive (though not unique) way of approaching their task of explaining the social and political world. Sometimes called **inference to the best explanation**, the distinctive logic of social scientific inquiry goes something like this:

> Given some data, and some candidate explanations or hypotheses that potentially explain that data, the explanation that is most compatible with the data is most likely to be true [1].

This abstract summary of the typical social science approach can be clarified by breaking it down into three concrete steps:

1. Generate possible explanations based on some initial observation(s)
2. Collect new data relevant to the implications of those explanations
3. Adjudicate between rival explanations in light of that new data

While these steps are the touchstone of most social science literature, the logic of inference to the best explanation is actually quite common in other avenues of life. Let's consider three non-social scientific examples.

1.1.1.1 Everyday reasoning

Figure 1.2: Wet morning grass

The first example consists of a rather mundane puzzle many of us know from our everyday lives. Perhaps you've had the experience of waking up in the morning, looking outside the window, and wondering: why's the grass all wet? Without exerting any extra effort, many of us slip right into inferring to the best explanation.

First, we generate possible explanations for observing wet grass in the morning:

- Overnight rain
- Morning dew
- Neighbor's sprinkler system

Next, we collect additional information based on the implications of those explanations:

- Check the weather app
  - Was rain in the forecast last night?
  - Was there high humidity yesterday?
- Look outside again
  - Is the sidewalk or the road also wet?

Finally, we critically assess what the new information implies about which explanation is more or less likely to be true. If the app says there was a 35% chance of rain last night *and* the road and sidewalk also appear wet, we gain confidence that overnight rain is the best explanation.

1.1.1.2 Mystery stories

Figure 1.3: Sherlock Holmes

A more explicit example of inference to the best explanation can be found in mystery stories, such as the novellas featuring famed fictional detective Sherlock Holmes or the many television crime and courtroom dramas found in the *Law & Order* extended universe. In instances such as these, the puzzle to be explained is, of course, who committed the crime?

Though the stakes are very different than when explaining wet grass in the morning, the logic of inquiry is basically the same.

- First, the investigator generates a list of potential suspects based on the initial facts of the crime.
  - *Who had the means, motive, and opportunity?*
- Next, the investigator collects additional evidence to corroborate the suspects' alibis.
  - *Are there credible witnesses that can verify their whereabouts on the night in question?*
- Finally, the investigator rules out suspects as likely perpetrators based on new evidence collected.
  - *Suspect A was out of town the during night in question so they couldn't have committed the crime.*

1.1.1.3 Medical diagnoses

Figure 1.4: The Pitt

HBO's hit medical drama *The Pitt* is set in a hospital emergency room in downtown Pittsburgh, Pennsylvania. Each episode is a one-hour increment of a 12-hour day shift, depicting in real time the stressful and chaotic environment in which overburdened and under-resourced hospital staff try to help their patients.

While each patient arrives with a distinct situation, the underlying logic of diagnosis is the same.

- Doctors generate a differential diagnosis based on patient's symptoms
  - *Are they presenting fever? Where do they report experiencing pain? Could it be X? Could Y be the cause?*
- Then doctors order lab tests and/or check patient for additional symptoms consistent with contending diagnoses
  - *What do the x-ray results show? Do their pupils respond normally to light?*
- Rule out diseases or injuries inconsistent with lab results and make a final diagnosis
  - *Based on this set of symptoms and a positive test result, the patient is most likely suffering from X*

As these examples illustrate, inference to the best explanation is a common form of reasoning found in diverse settings outside of social science. But it is also foundational to the work of social scientists. Let's consider an example.

1.1.1.4 Political science: explaining the rise of right wing populism

Figure 1.5: Right wing populists: (from left to right) Donald Trump, Giorgia Meloni, Viktor Orbán, Nigel Farage

One of the major debates currently animating political science concerns explaining the rise of right wing populism across the world over the last few decades [2]. Despite the clear differences from everyday life, murder mysteries, and emergency rooms, we can see inference to the best explanation at work in this field as well.

- Researchers theorize potential explanations based on initial observation of rising right wing populism
  - *Economic grievance*
  - *Sociocultural grievance*
  - *Government dysfunction*
  - *Entrepreneurial politicians*
- Next, researchers collect data to test implications of rival explanations
  - *Does low socioeconomic status increase voters' support for populist candidates?*
  - *Is there a cross-country relationship between anti-immigrant attitudes and populist electoral success?*
  - *Is there growing citizen dissatisfaction with political institutions?*
  - *How do establishment politicians respond to the challenge of populist insurgents?*
- Adjudicate between rival explanations in light of new data
  - *Which explanation (or combination of explanatory factors) is most compatible with the data?*

Let's pause here to recognize how this example returns us to the description of social science research offered at the beginning of this guide. While we've layered on the label **inference to the best explanation** in this chapter, we are only re-describing the research process in terms of the particular mode of social science reasoning happening under the surface.

It begins with an initial observation—wet grass, a robbery, a feverish patient, or the election of a populist—which presents us with a **problem or puzzle** in need of explanation. Having made this initial observation, we generate potential explanations or **theories** on our own or more often in **conversation** with other specialists. To assess which theory is most compatible with the data, we **gather additional information** by testing the theories' empirical implications. *If theory A is true, what would I expect to see* *in the world?* The aggregate weight of the evidence collected in **hypothesis testing** helps us infer which theory or explanation is most likely to be true.

1.2 Strengths and limitations of social science research

Inference to the best explanation is the underlying logic of social science and is explicitly exhibited in the best social science work. This has clear virtues

1.3 The research project workflow

Let's start by taking a bird's eye view of this semester's research project as a whole. Figure 1.6 displays a diagram representing the research workflow, starting and ending with the state of existing knowledge.

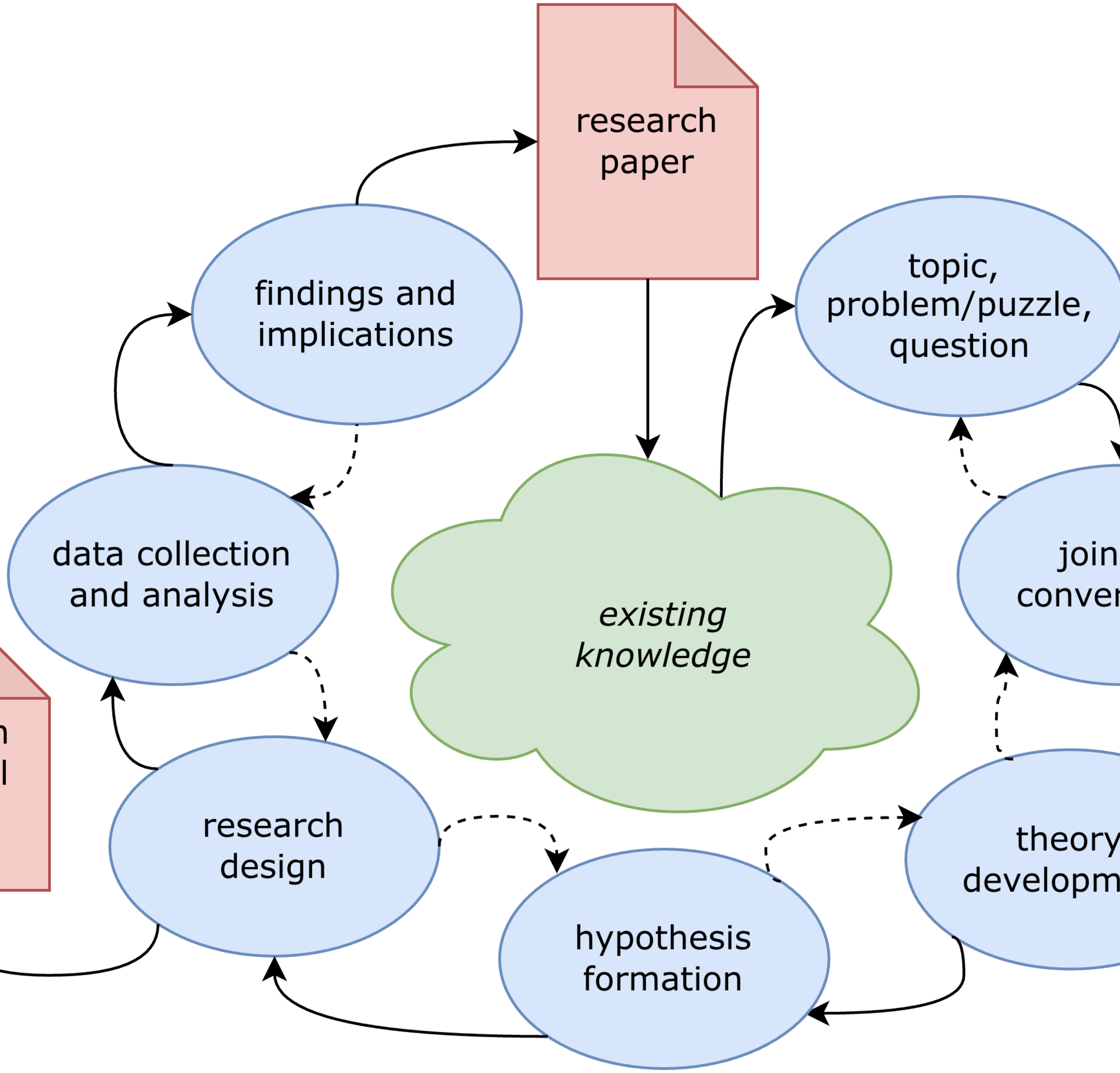

Figure 1.6: The research workflow

There's a lot going on in this diagram, so let's explain it piece by piece. The blue-shaded bubbles represent distinct stages of the research process. The solid arrows trace a clockwise workflow from **topic** in the top-right corner to **findings and implications** near the top-left. The dashed arrows trace a counter-clockwise workflow back through the stages. This is meant to capture that although the workflow looks chronological, real-world research typically iterates back and forth between stages, recursively revising and refining previous stages as we move forward to new ones.

There are also two red-shaded deliverables that are spun out of the workflow at various points. A formal **research proposal** is expected around the midpoint in the semester and a polished **research paper** is expected at the end.

As the semester proceeds, we'll spend significant time unpacking each and every one of these stages as you move your project through the research workflow. I'll also clarify precisely what is expected in the research proposal and paper. The chapters that follow do precisely that. For now, to get a better sense of the big picture, let's take a brief tour through the workflow.

Existing knowledge

All research projects begin and end with the state of existing knowledge. We draw from it when we learn about a subject that interests us, and we contribute to it when we communicate our research findings to audiences and stakeholders. As researchers, our intention is to extend the frontiers of knowledge, even if only modestly, leaving the field in a better place than when we found it.

Topic, problem/puzzle, question

While existing knowledge typically plays an implicit, background role, in the formation of research projects, specific topics, problems, puzzles, or questions are usually top of mind in the formation of research projects.

Join the conversation

Theory and hypotheses

Research design

Data collection and analysis

Findings and implications

1.3.1 Inference is the goal

There's a common misconception among new students in the social sciences that they will find a single correct answer to their research question—something along the lines of, say, $2 + 2 = 4$.

Inference is simply the act of making claims that go beyond the particular evidence examined [3].

But if by *answer* we mean a transparently reasoned, evidence-based conclusion that is amenable to future updating based on new evidence, then let me reassure you that you're ready for the challenge that awaits you.

The process by which we arrive at this transparently reasoned, evidence-based, and always-tentative conclusion goes by the name **inference**, and it is the basis of all social science research. According to its Wikipedia entry, inference is simply a "conclusion reached on the basis of evidence and reasoning." Why must we infer?

We make so many inferences in our everyday lives, we may not even consciously notice. When we wake up and find that the sidewalk is wet, we infer that it rained overnight. When you see a friend sitting down to have a snack, you may infer that they're hungry. If you find your roommate packing a bag, you might infer that they are going away for the weekend.

In the social sciences, we make inferences about other kinds of phenomena, but the logic is the same. After observing a polity's electoral process, we may classify the regime as either a democracy or an autocracy. To survey public opinion, we typically rely on a representative sample from which we generalize about the whole population. If the economy enters a recession, we may expect an incumbent's chances of reelection to diminish.

These examples of inference from both everyday life and the social sciences have something in common: they all occur under conditions of **uncertainty**. This is why social science research is fundamentally unlike concluding that $2 + 2 = 4$. The simple definition of the equation's symbols provides us with complete information and makes the conclusion logically inescapable. We can claim that $2 + 2 = 4$ with absolute certainty.

When it comes to social science, we not so lucky. Typically, we operate with incomplete information and must form our conclusions carefully to account for uncertainty. We may think that it was the economic recession that doomed the president's reelection, but perhaps it was a corruption scandal that was top of mind for voters that year. Needing to do our research under conditions of uncertainty makes social science challenging but also interesting and ultimately rewarding.

1.3.2 Inference and explanation

The centrality of inference to social science research may feel a little deflating at first. It is natural to want to find clear and definite answers to questions of widespread public concern and political importance. Does the ubiquity of uncertainty mean that we can't really explain anything?

Why do much uncertainty? There are multiple sources of uncertainty when conducting research on the social and political world [4, p. 73].

1.3.3 A pluralist methodology

I'll conclude this introductory chapter with a brief statement intended to clarify the methodological underpinnings of the research advice contained in this book. Social science can be thought of as a mansion with many rooms,

Philosophically, this book embraces what's called methodological pluralism. This particular view of social science follows from the principles laid out above. If we as researchers operate under conditions of uncertainty and our inferences are always tentative and subject to revision, then it's best that we take a broadly inclusive approach to social science research. Gatekeeping—that is, judging what is and *is not* legitimate social science—can be a harmful and unproductive practice, and is

This is not the same as saying that social science is a field in which "anything goes." As we've established, there are standards that must be met.

2. From Topic to Question

If social scientists' ultimate goal is to explain the social and political world, it's unsurprising that research projects typically spring from a problem or puzzle that initially defies explanation.

3. Joining the Conversation

4. Theory and Hypotheses

4.1 Introduction

Whether we consciously realize it or not, we humans engage with our everyday surroundings by constantly constructing, deploying, and revising mental models of how the world works. These models may be flawed, incomplete, or even downright wrong. Nevertheless, so long as they prove useful for our particular purpose, we continue to use them.

Social scientists concerned with explaining how aspects of the social and political world works build models too. Unlike most of the mental models we walk around with unconsciously, social scientists must make their models explicit by articulating **theories**, expressed in the form of either words, diagrams, mathematics, or all of the above.

Simply put, theories are just relationships between concepts, linked together by a set of assumptions made explicit by whomever is building the model. As such, theories are neither true nor false in themselves —no more so than a painting or a subway map is true or false. A model is simply a representation of reality, designed with a particular purpose in mind [5]. A Picasso-style portrait, though visually appealing, will not work for a passport photo. Nor will a subway map be very useful if you are navigating a city on foot.

<Picasso and subway map images side by side>

The usefulness of any particularly theory is found in how well it helps us explain the empirical phenomena it predicts. Which theory is more likely given the evidence?

In short, a theory is a proposed answer to a research question. Theory development is the process of "identifying numerous potential explanations [to your question] and then sifting through them to find the one that you feel is the most likely or strongest explanation of the phenomenon" [6, p. 34].

We address theory development and hypothesis formation together in this chapter because they so often go hand-in-hand during actual research.

4.2 Visualizing theory with DAGs

5. Research Design

6. Research Proposal

7. From Proposal to Paper

8. Data Collection and Analysis

8.1 Common Data Resources in Political Science

This section (borrowed from L. C. Powner [6]) contains an extensive collection of dataset links. If you can imagine it, there's a VERY good chance someone else has already collected data on it, and one of these sites has a link to it. If you're struggling to find appropriate data, let me know.

8.1.1 University Libraries

https://guides.libraries.emory.edu/main/Data_Services/govinstlinks (Also their US States page)

https://guides.lib.umich.edu/?b=s Hundreds of field-specific guides to research data; use the Search tool for easiest access

https://libguides.princeton.edu/politics/data – good collection of data for beginners

8.1.2 Cross-Subfield

www.poliscidata.com Barry Edwards has an INCREDIBLE collection of data sets from across the field, organized by subfield and topic. If you know what you want, this is the place to go.

https://dataverse.harvard.edu/ Data archive. Contains both article data (often organized by publication name) and archives for larger or longer-running data collection programs.

https://datasetsearch.research.google.com/ Google has a dataset search tool. It's NOT my favorite, but if you are looking for a unique piece of data or something from outside conventional political science themes, this may be helpful. (Recommended for more advanced users.)

Dataset with Political Datasets https://github.com/erikgahner/PolData Databases catalogued by content and links. Collection of cross-national election studies, cabinet data, political institutions, constitutions, parties & politicians, governance, economics, and more

8.1.3 American Politics

https://electionstudies.org/ – The American National Election Studies datasets are the gold standard for US political behavior, with 75 years of data and nearly 10,000 citations. Most people start with the question search tool. Much data available for political psychology, race & politics, and attitudes towards public figures, groups and issues.

Cooperative Election Study (Formerly Cooperative Congressional Election Study)

https://tischcollege.tufts.edu/research-faculty/research-centers/cooperative-election-study Starting in 2006, the study captures large numbers of Americans (think 50,000 respondents a year) before and after each national election. Besides a common content module asked of all respondents, teams of researchers add additional thematic modules that typically get asked of 1,000-2,000 respondents. Many unique theme modules thus exist in this data, though it's a bit more difficult to navigate.

State Politics & Policy Quarterly data resource https://web.archive.org/web/20150311202414/https://academic.udayton.edu/SPPQ-TPR/tpr_data_sets.html

Correlates of State Policy https://ippsr.msu.edu/public-policy/correlates-state-policy State policy adoptions and changes over time, plus associated state-level variables

State Ideology Data: https://rcfording.wordpress.com/state-ideology-data/

8.1.4 Non-US and Comparative Politics

Varieties of Democracy (V-Dem) https://www.v-dem.net/ WAY more than just democracy – all sorts of political and economic institution data in one place. Often the first stop for building cross-national datasets.

Polity https://www.systemicpeace.org/ The baseline institutional measure of democracy for decades. Data stop around 2020 for technical reasons.

Comparative Agendas Project https://www.comparativeagendas.net/ What are the hot political issues in various countries and US states over time? Only a few US states have their own data, but lots of cool country-based data.

Database of Political Institutions (Interamerican Development Bank) https://data.iadb.org/dataset/the-database-of-political-institutions-dpi-2020 Publicly accessible, 45 years, 180 countries

https://github.com/erikgahner/PolData Collection of cross-national election studies, cabinet data, political institutions, constitutions, parties & politicians, governance, economics, and more

Comparative Study of Electoral Systems https://cses.org/ National electoral studies designed to be comparable. Includes most moderately or fully democratic countries.

National Elections Across Democracy and Autocracy (NELDA) http://nelda.co/ Data organized by election round for all recognized countries in the world. Data access by (very simple) request.

8.1.5 International Relations

THOUSANDS of specialized IR datasets exist. https://github.com/erikgahner/PolData and Poliscidata.com catalog many titles thematically.

Design of International Trade Agreements (DESTA) https://www.designoftradeagreements.org/downloads/ Treaties list, withdrawals, agreement content

Alliance Treaty Obligations Project (ATOP) http://www.atopdata.org/ All military alliances formed between 1815-2018 along with their provisions; [NEWGENE replacement] allows easy combination with conflict data

Cignarelli-Richardson Human Rights Data (CIRI) – https://dataverse.harvard.edu/dataverse/cirihumanrightsdata The most respected non-governmental measurement of human rights

Correlates of War project (COW) https://correlatesofwar.org/ best known for war data, militarized interstate disputes (MIDs), and national material capabilities, but many other security related topics are available (contiguity, defense cooperation, shared international organization membership, etc.)

Global Terrorism Database https://www.start.umd.edu/data-tools/GTD domestic & transnational terrorism

International Crisis Behavior Events – https://www.crisisevents.org/ crises smaller than wars or militarized interstate disputes Social Conflict Analysis Database (SCAD) – Africa & Latin America/Caribbean; riots, strikes, protests, government violence against civilians.

9. Findings and Implications

10. Research Paper

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